

Unit 2 — Exponent Rules

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

§1 Product & Quotient. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; “Same base? ADD.” Quotient: $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; “Same base? SUBTRACT.”

§2 Power. Power: $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; “Power to power? MULTIPLY.” Power of a Product: $(2x)^3 = 8x^3$; $(3ab)^2 = 9a^2b^2$; “Give EACH factor the power.” Power of a Quotient: $(2/3)^2 = \frac{4}{9}$; $(a/b)^3 = \frac{a^3}{b^3}$; “Power to top AND bottom.” Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

§3 Zero & Negative. Zero: $5^0 = 1$; $(3x^5)^0 = 1$; “ANY base⁰ = 1.” Negative: $2^{-3} = \frac{1}{8}$; $4x^{-3} = \frac{4}{x^3}$; “Flip to a fraction.” **Across the bar:** $\frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$. **Reciprocal:** $(\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$.

Practice 1 — Product & Quotient

1) x^{11} 2) $15x^8$ 3) $8a^5b^7$ 4) $-18x^5y^4$ 5) $30m^3n^6$ 6) x^5 7) $5a^5$ 8) $4x^3y^4$ 9) $4m^5n^2$ 10) $-6p^4q^3$ 11) $3x^5$ 12) $2a^4b^2$

Practice 2 — Power Rules

1) x^{20} 2) $16x^{12}$ 3) $-27a^6$ 4) x^6y^{15} 5) $16m^6n^4$ 6) $32x^{15}y^{20}$ 7) $\frac{x^8}{y^{12}}$ 8) $\frac{8a^3}{b^6}$ 9) $\frac{9x^8}{4y^2}$ 10) x^{24} 11) $72x^8y^7$ 12) $16a^{12}b^8$

Practice 3 — Zero & Negative

1) 1 2) 1 3) $\frac{1}{64}$ 4) $\frac{2}{x^5}$ 5) $\frac{y^2}{x^3}$ 6) $\frac{b^3c}{a^2}$ 7) $\frac{4n^5}{m^6}$ 8) $\frac{4y^3}{x^4}$ 9) $\frac{y^2}{x^2}$ 10) $\frac{27}{8}$ 11) $\frac{b^3}{a^6}$ 12) $\frac{y^4}{9x^2}$

Cumulative Review

A • Exponent Rules: 1) x^{10} 2) $12x^7$ 3) $10a^4b^6$ 4) $4x^4$ 5) m^{15} 6) $16x^8y^{12}$ 7) $6a^3b^4$ 8) $\frac{x^{12}}{y^6}$

B • Negative Exponents: 1) $\frac{1}{25}$ 2) $\frac{3}{x^4}$ 3) $\frac{y^3}{x^5}$ 4) $\frac{b^3}{a^3}$ 5) $\frac{5b^6}{a^5}$ 6) $\frac{y^2}{4x^4}$